

Project Proposal

1. Project Title

Design, Implementation and Verification of a 4-Bit Arithmetic Logic Unit (ALU) Using Cadence Virtuoso

Course / Section	Track	Submission
(Course code and section)	A: 4-Bit ALU (datapath component)	(10 October 2026)

2. Introduction

A **datapath** is the part of a digital processor that stores, moves and transforms data. It is built from registers, multiplexers and an arithmetic logic unit (ALU), and it is steered by control signals produced by a control unit. Datapath design matters in digital and VLSI systems because it sets the speed, area and power of the whole processor: the longest path through the ALU usually decides the maximum clock frequency.

This project implements the **ALU**, the computational core of a datapath. It takes two 4-bit operands, A and B, and a 2-bit opcode, and returns a 4-bit result plus carry-out and overflow flags. The design is built hierarchically (gates, full adder, 1-bit slice, 4-bit ALU) in Cadence Virtuoso, with a symbol and testbench at every level so that each block can be verified on its own before integration.

3. Problem Statement

Digital systems need a compact combinational unit that can perform arithmetic and logical operations on small words under the control of select signals. The problem is to design a 4-bit ALU that supports at least addition, subtraction, AND and OR (with optional extensions), produces correct carry and overflow flags, is fully verified by simulation for all operations including boundary cases, and is structured so that it can later be connected to a register set, a 16x4 memory multiplexer and a control unit to form a complete 4-bit datapath.

4. Objectives

- Design and verify the basic gates (NOT, AND, OR, XOR) and a 1-bit full adder at schematic level.
- Design a 1-bit ALU slice combining an adder/subtractor, AND, OR and a 4:1 output multiplexer.
- Chain four slices with a ripple-carry path to form the 4-bit ALU and generate Cout and overflow flags.
- Create a symbol for each level of the hierarchy so the ALU can be reused in a larger datapath.
- Write testbenches, simulate in Virtuoso ADE, and compare expected and actual outputs for every opcode.
- Measure functional correctness and worst-case propagation delay (carry ripple path).
- Extension (if time permits): add XOR, XNOR, shift-left, shift-right and comparison using a 3-bit opcode.

5. Proposed System

5.1 Operation table. A 2-bit opcode selects the operation.

Op[1:0]	Operation	Result	Sub
00	ADD	$R = A + B$	0
01	SUBTRACT	$R = A - B (A + \text{NOT } B + 1)$	1
10	AND	$R = A \text{ AND } B \text{ (bitwise)}$	0
11	OR	$R = A \text{ OR } B \text{ (bitwise)}$	0

5.2 Output equations (for bit $i = 0$ to 3; $C_0 = \text{Sub}$).

$$\text{Sub} = \text{Op}1' \cdot \text{Op}0$$

$$B'_i = B_i \text{ XOR } \text{Sub}$$

$$S_i = A_i \text{ XOR } B'_i \text{ XOR } C_i$$

$$C_{i+1} = A_i \cdot B'_i + C_i \cdot (A_i \text{ XOR } B'_i)$$

$$R_i = \text{Op}1' \cdot S_i + \text{Op}1 \cdot \text{Op}0' \cdot (A_i \cdot B_i) + \text{Op}1 \cdot \text{Op}0 \cdot (A_i + B_i)$$

$$\text{Cout} = C_4$$

$$\text{Overflow } V = C_4 \text{ XOR } C_3 \text{ (meaningful for ADD and SUBTRACT only)}$$

5.3 Architecture. Figure 1 shows the top level: an opcode decoder generates Sub, and four identical 1-bit slices are connected by a ripple-carry chain. Figure 2 shows one slice: an XOR gate conditionally inverts B, a full adder produces the sum, AND and OR gates run in parallel, and a 4:1 multiplexer driven by Op[1:0] selects the result bit.

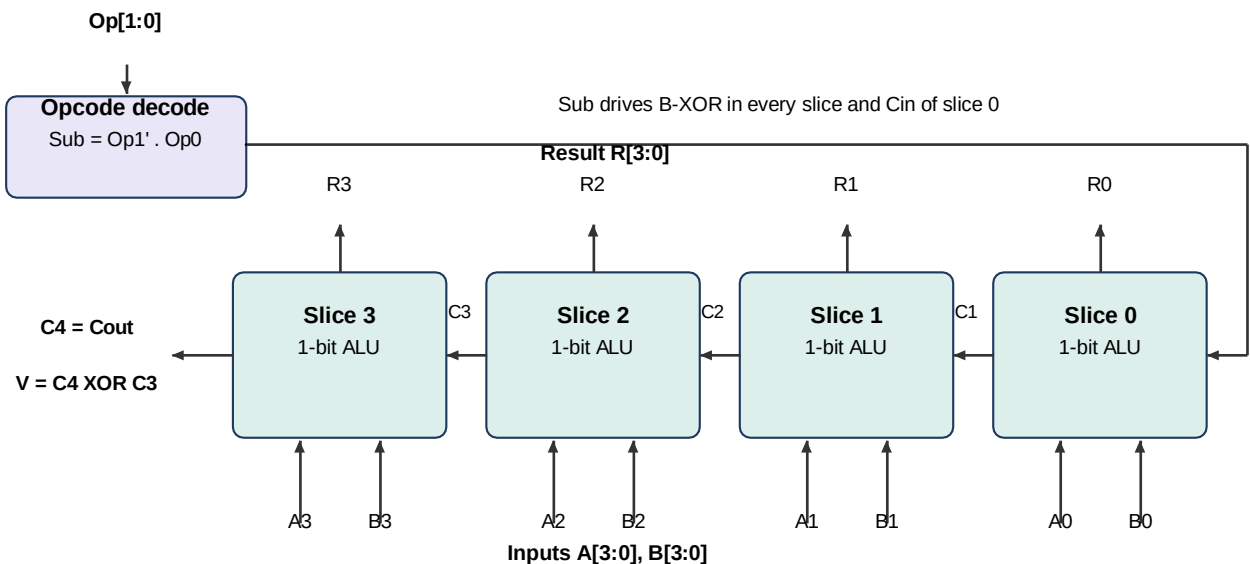


Figure 1. Top-level block diagram of the 4-bit ALU

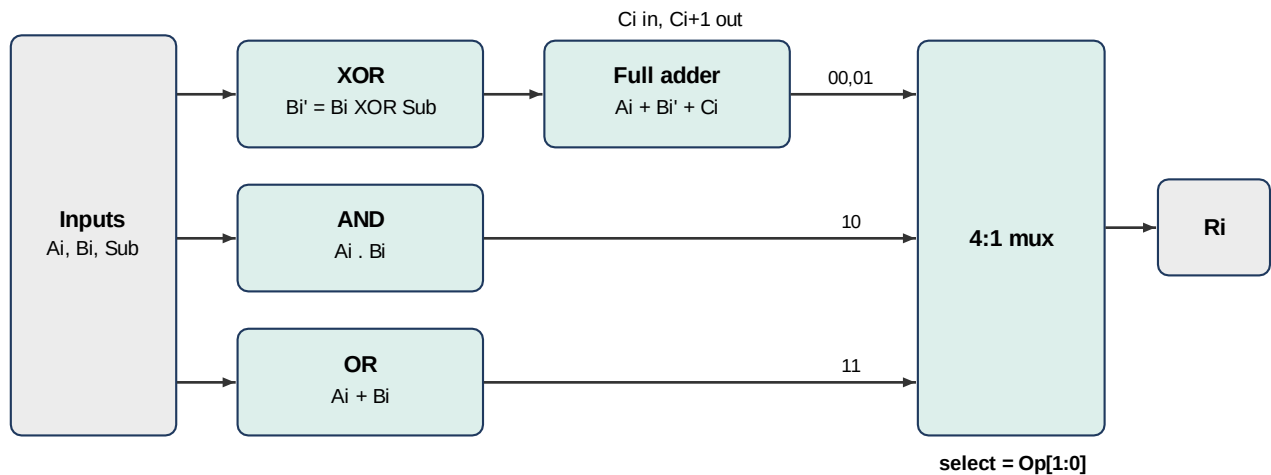


Figure 2. Internal structure of one 1-bit ALU slice

6. Tools and Technologies

Tool / Technology	Purpose
Cadence Virtuoso Schematic Editor	Gate-level and hierarchical schematic entry, symbol creation
Cadence ADE (Spectre simulator)	Transient simulation, waveform viewing, delay measurement
Process design kit (e.g. GPDK 180 nm)	Standard cells / transistor models used by the schematics
Verilog HDL + simulator (optional)	Behavioural reference model to cross-check expected outputs
Word/LaTeX and screenshots	Final report with schematics, symbols and waveforms

7. Expected Outcome

- Complete 4-bit ALU schematic with 1-bit slice, full adder and gate sub-blocks.
- Symbols for each hierarchy level.
- Testbenches for the full adder, the 1-bit slice and the 4-bit ALU.
- Simulation waveforms for all opcodes, with verified outputs and flags.
- Block diagrams and output equations (Section 5).
- Propagation-delay measurement of the worst-case carry path.
- Synthesis results from the optional Verilog model, if required.
- Final project report.

8. Testing and Verification Plan

Module-level testing. The full adder is tested with all 8 input combinations. The 1-bit slice is tested with all combinations of A_i , B_i , C_i and the four opcodes. The 4-bit ALU is then tested with the directed cases below, plus a sweep of all 256 (A, B) pairs for ADD and SUBTRACT if simulation time allows. Expected values are computed by hand or by the Verilog reference model and compared with the simulated outputs.

Case	Op	A	B	Expected R	Cout	V
Normal add	00	0101	0011	1000	0	1
Add with carry out	00	1111	0001	0000	1	0
Zero + zero	00	0000	0000	0000	0	0
Normal subtract	01	0101	0011	0010	1	0
Subtract to negative	01	0011	0101	1110	0	0
Subtract equal	01	0111	0111	0000	1	0
AND	10	1100	1010	1000	-	-
OR	11	1100	1010	1110	-	-
Boundary: max values	00	1111	1111	1110	1	0
Boundary: signed overflow	00	0111	0001	1000	0	1

Notes. Cout for subtraction is 1 when no borrow occurs ($A \geq B$). Overflow is interpreted for two's-complement values. All four opcodes are exercised with the same operands to confirm that the multiplexer selects correctly. The ALU is purely combinational, so no clock or reset is needed; if the ALU is later registered inside the datapath, clock and reset behaviour will be verified at integration time. Worst-case delay is measured by switching A0 or Sub so the carry travels from slice 0 to slice 3.

9. Team Responsibility

Member (Name, ID)	Responsibility
Abdullah Al Jubayer (22201245)	Basic gates, full adder schematic and testbench
H. M. Tahsin Sheikh (22201243)	1-bit slice, 4:1 mux and opcode decoder
Shahrier Shimul (22201227)	4-bit integration, symbols, flags (Cout, V), delay measurement
All members	Simulation, verification tables, final report and presentation